Hassan El Khatoury

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LinkedIn · GitHub · Portfolio

Professional Summary

Computer Science Engineering student specializing in **Java backend development**, distributed systems, and scalable architectures. Dual **Oracle Certified Professional Java Developer (Java SE 11 & 17)** and OCI Foundations certified. Experience building backend services, REST APIs, and distributed database architectures. Seeking a **Software Engineering Internship (Backend / Java)**.

Education

ENSAH — National School of Applied Sciences, Al Hoceima 2022–2027
Engineering Degree in Computer Science (Engineering Cycle)
Relevant Coursework: Algorithms, Advanced Java, Software Architecture, Databases, Computer Networks, Linux

Baccalauréat in Mathematical Sciences (French Option) 2022

Technical Skills

Languages: Java, SQL, Python, C/C++, JavaScript, TypeScript
Frameworks: Spring Boot, FastAPI, React, Next.js, JavaFX
Databases: Oracle Database, MySQL, MongoDB (Sharding, Replication), PL/SQL
Cloud & Tools: Oracle Cloud Infrastructure, Git, GitHub, Linux, Maven, Docker
Concepts: Microservices, REST APIs, OOP, Design Patterns, Distributed Systems, Data Structures & Algorithms
AI / Data: Machine Learning, Scikit-learn

Experience

Web Development Intern — CodSoft (Remote) Feb 2025 – Mar 2025

- Developed and shipped **10+ features** across full-stack web applications using REST APIs
- Collaborated using Git workflows, pull requests, and version control best practices
- Applied modular architecture principles improving maintainability and code quality

Projects

InterviewAI — Java Desktop Application

- Designed layered architecture with **BCrypt authentication** supporting multi-user workflows
- Integrated REST APIs and optimized MySQL data access using JDBC
- Built admin analytics dashboard and gamification system to improve engagement
- *Tech: Java 17, JavaFX, Maven, MySQL, REST APIs*

MongoDB Sharded Cluster Lab — Distributed Database Deployment

- Deployed a distributed MongoDB cluster using Docker with replica sets and shard servers
- Implemented sharding to distribute transport datasets across multiple nodes
- Configured replication and failover to ensure high availability
- Demonstrated distributed database architecture including config servers, routers, and shard nodes
- *Tech: MongoDB, Docker, Distributed Systems*

CAN-2025 Predictor — Machine Learning System

- Built Random Forest model achieving **∼70% accuracy** on historical match data
- Developed FastAPI backend serving predictions to a Next.js tournament simulator
- Implemented Monte Carlo simulations (**10,000+ runs**) to forecast competition outcomes
- *Tech: Python, FastAPI, Scikit-learn, Next.js*

Certifications

- Oracle Certified Professional: Java SE 17 Developer
- Oracle Certified Professional: Java SE 11 Developer
- Oracle Cloud Infrastructure 2025 Foundations Associate
- Git & GitHub Fundamentals for Developers

Languages

Arabic (Native) · English (Fluent) · French (Intermediate)